

C++ Basics

1. What Is C++ and Where Is It Used?
2. Your First C++ Program
3. Variables and Data Types
4. Getting User Input and Displaying Output
5. Operators in C++
6. Making Decisions with `if` and `else`
7. Repeating Actions with Loops
8. Writing and Using Functions
9. Arrays and Vectors
10. Working with Strings
11. Introduction to Pointers and References
12. Introduction to Classes and Objects
13. Common Programming Errors and Debugging
14. How to Write Clear, Readable Code
15. Building a Simple Calculator in C++

General Programming

16. How Computers Follow Instructions
17. What Are Algorithms? Everyday Examples
18. Flowcharts and Pseudocode
19. How Programming Languages Differ
20. Introduction to Data Structures
21. What Is Version Control? An Introduction to Git
22. How to Plan and Build a Small Software Project
23. How a Program Works: From Code to Output
24. Choosing the Right Programming Language
25. Algorithms in Everyday Life (*)
26. Flowcharts and Pseudocode (***)
27. Functions: Reusing Code (*)
28. How to Find and Fix Bugs (*)
29. Writing Readable and Well-Organized Code
30. How Games Are Programmed
31. How Websites and Apps Are Built
32. Introduction to Data Structures
33. Object-Oriented Programming for Beginners
34. How APIs Let Programs Communicate
35. Version Control and Git
36. Testing Software Before Release
37. Cybersecurity Basics for Programmers
38. Artificial Intelligence and Programming
39. Programming Robots to Respond to Sensors
40. The Software Development Life Cycle
41. Open-Source Software and Collaboration

- 42. How Computers Represent Text, Images, and Sound
- 43. Why Programs Need Error Handling
- 44. Building a Simple Project from Idea to Prototype
- 45. The Future of Programming
- 46.

Artificial Intelligence

- 1. How Chatbots and Large Language Models Work
- 2. Generative AI: Creating Text, Images, and Music
- 3. How C++ Can Be Used in AI and Robotics
- 4. Vibe Coding: Turning Ideas into Apps with Natural Language
- 5. How Google Antigravity Helps Build Software
- 6. From Prompt to Prototype: Create a Small App with AI
- 7. How to Write Clear Prompts for Coding Tools
- 8. AI Coding Agents: How They Plan, Code, and Check Their Work
- 9. Vibe Coding vs. Traditional Programming
- 10. Debugging Code Generated by AI
- 11. Why Programmers Still Need to Understand Code